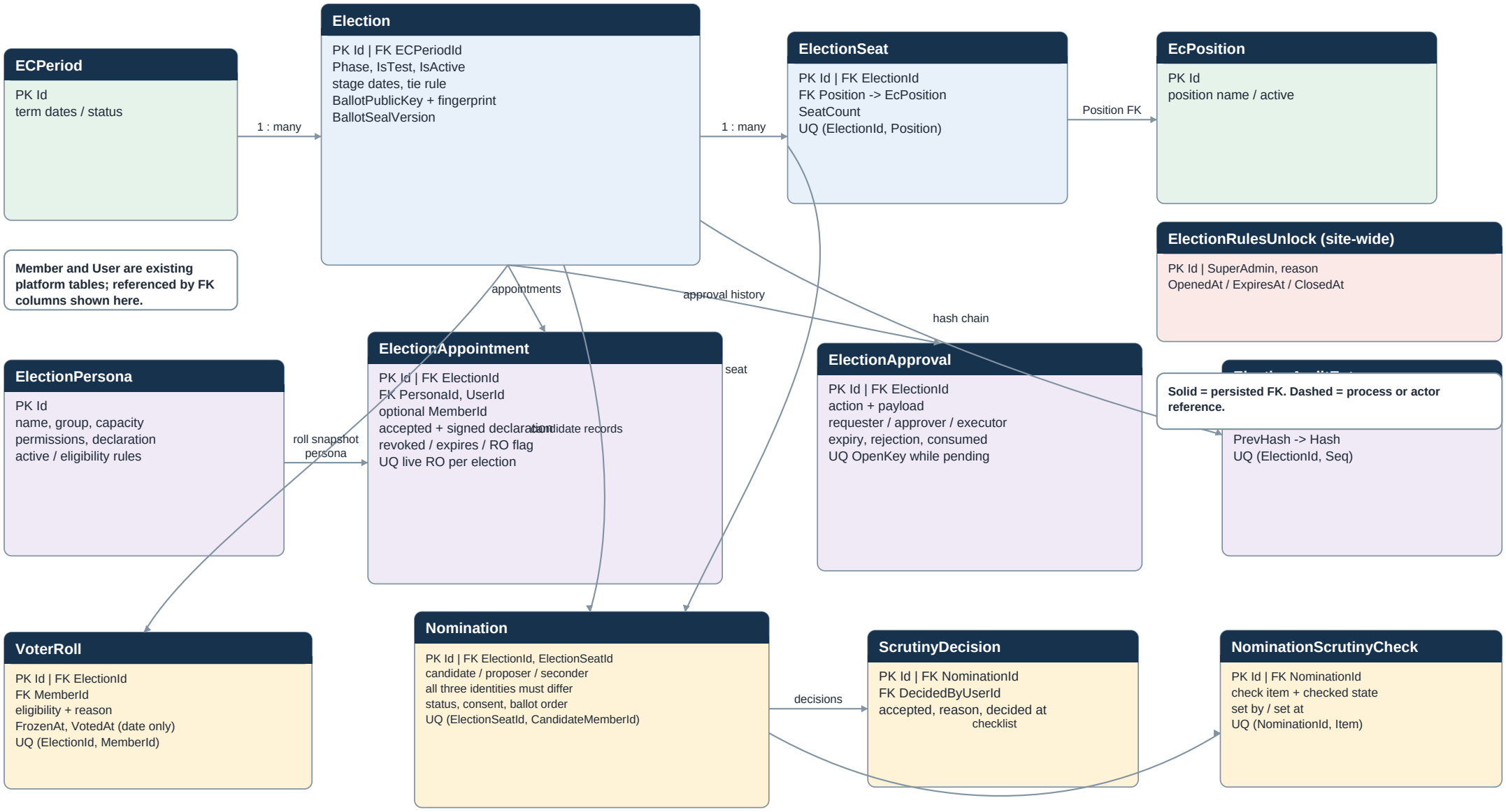
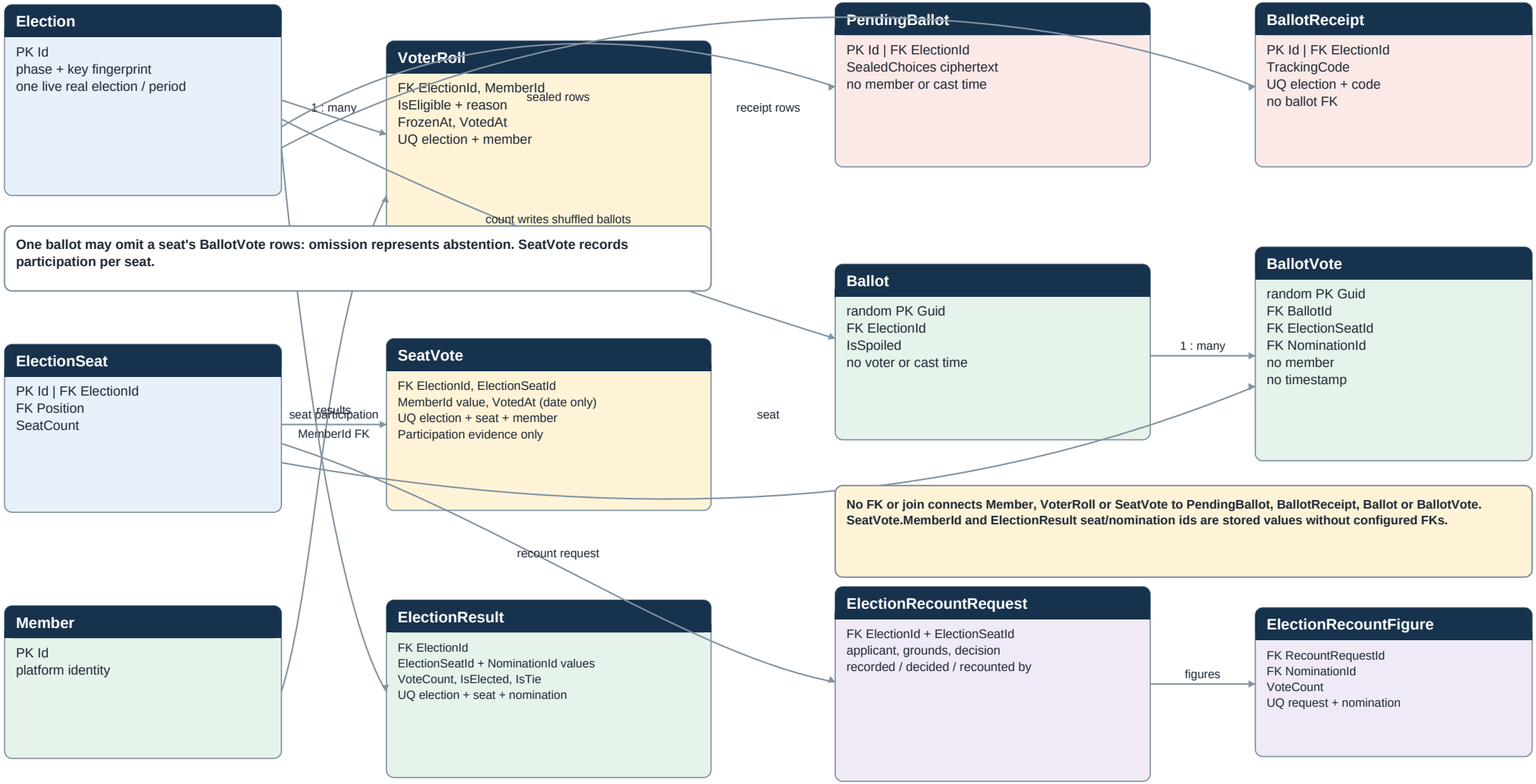
Election ERD | 1. Governance, eligibility and nominations

Persisted election setup and candidate preparation. PK/FK labels reflect current domain and EF configuration.



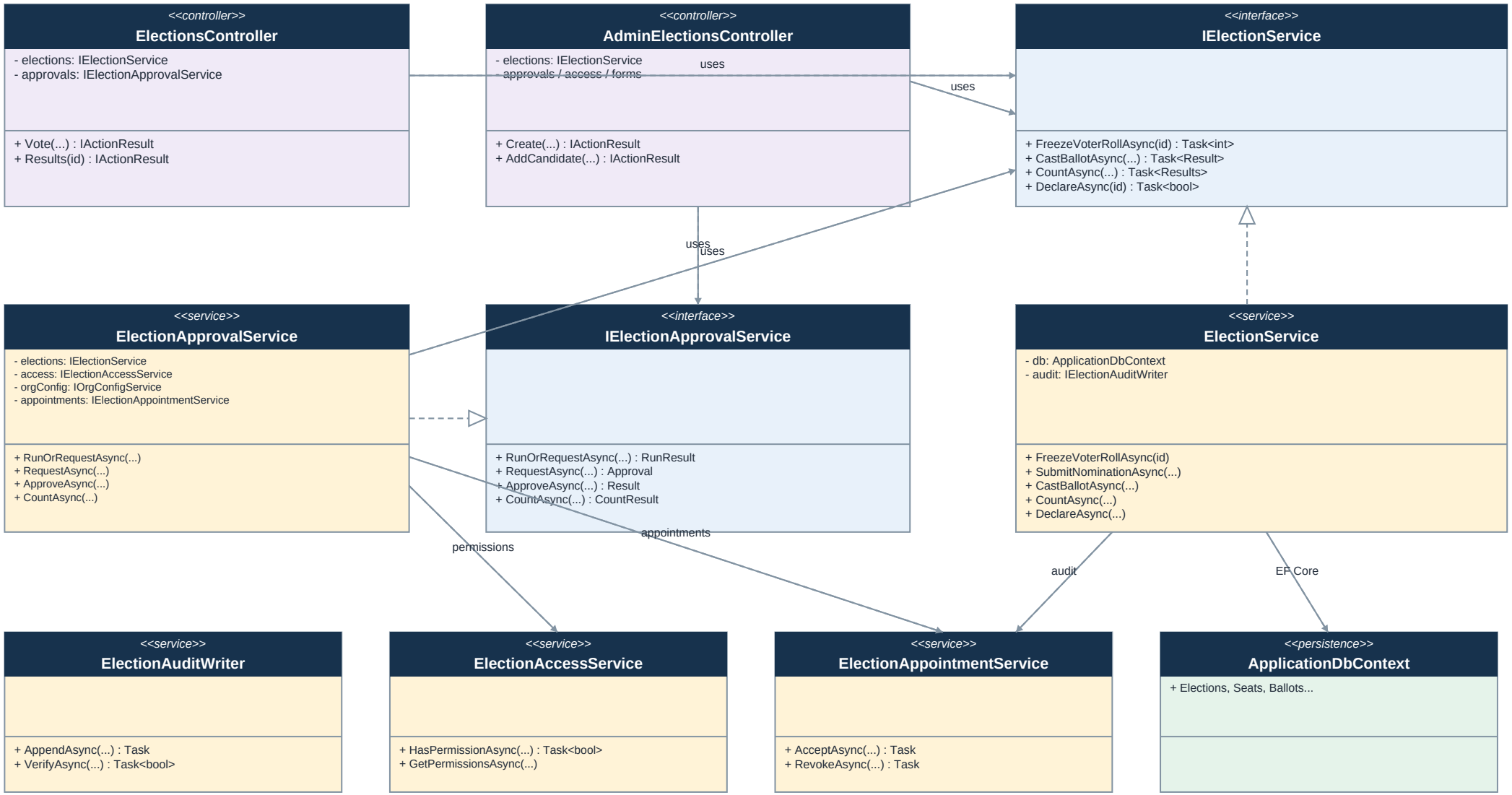
Election ERD | 2. Participation, sealed ballots and outcomes

Ballot secrecy depends on keeping voter-side evidence separate from anonymous counted choices.



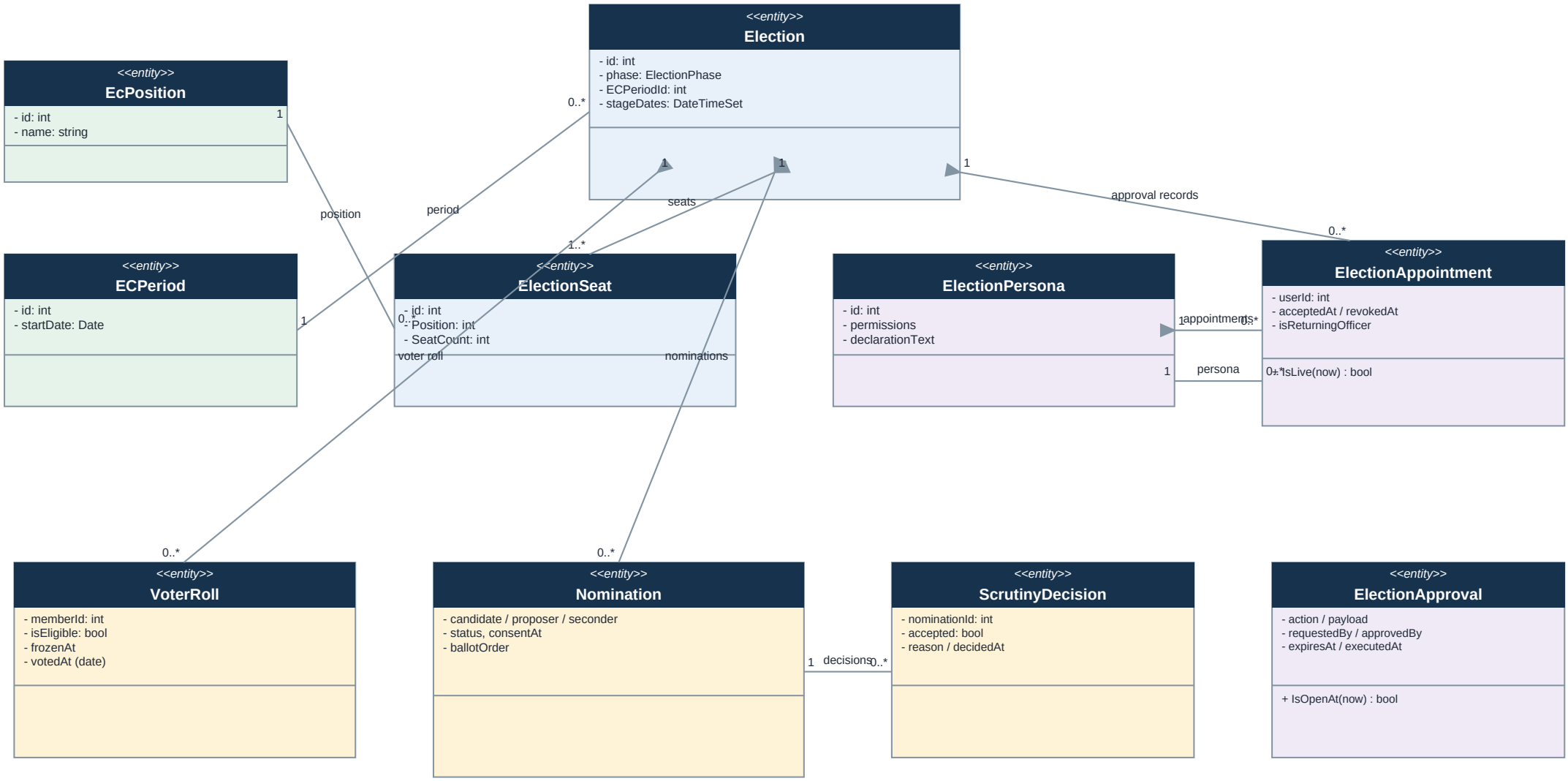
UML class diagram | API and application services

Class compartments show visibility, key fields and operations. Dashed realization implements an interface; dashed arrows are dependencies. Client UI is omitted.



UML class diagram | Election governance and nominations

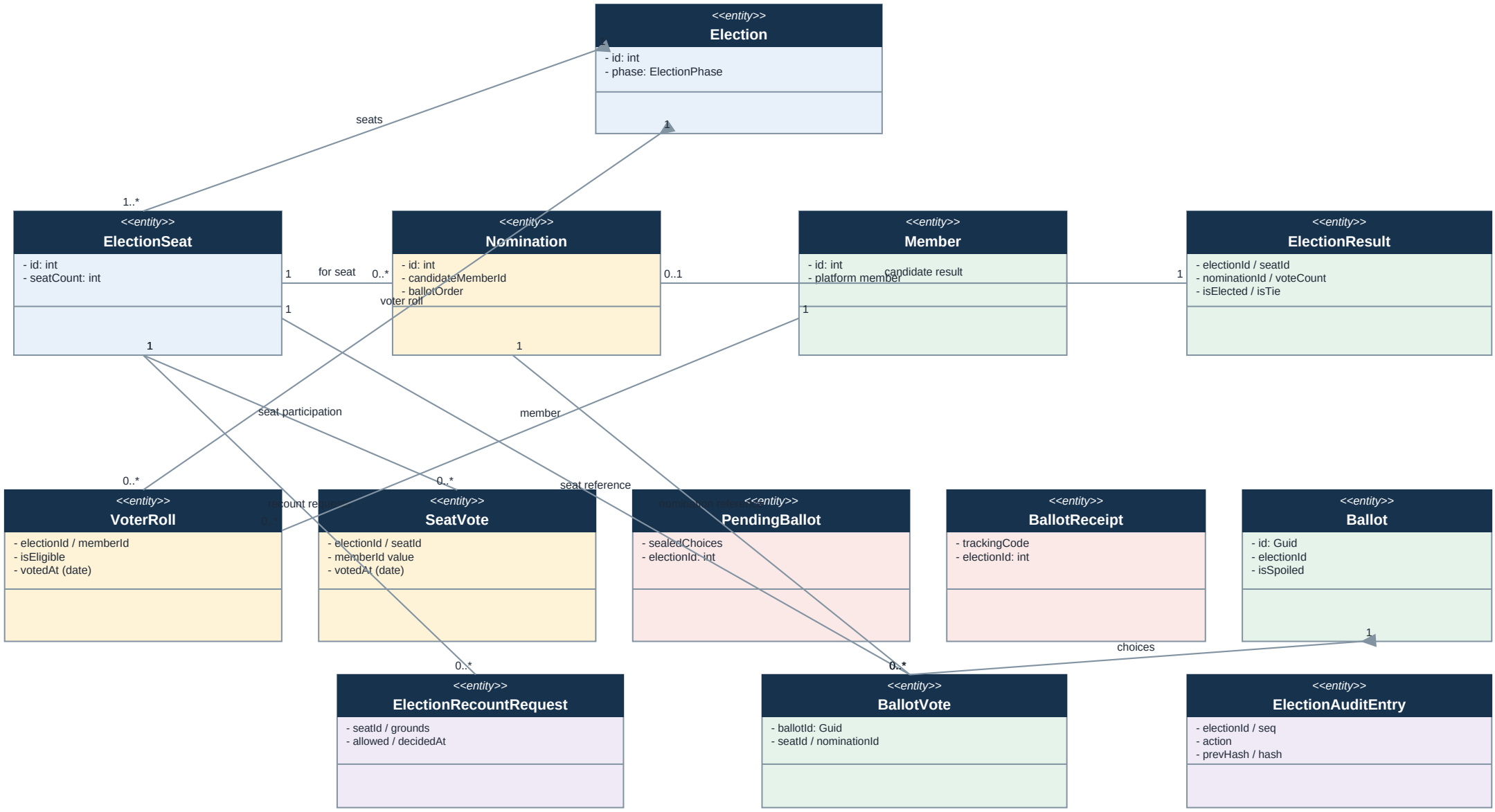
Domain classes with key attributes, behavior and association multiplicities.



Multiplicity: 1 = exactly one; 0..* = zero or more; 1..* = one or more. Candidate, proposer and seconder differ. ElectionRulesUnlock is site-wide. ERDs show database constraints.

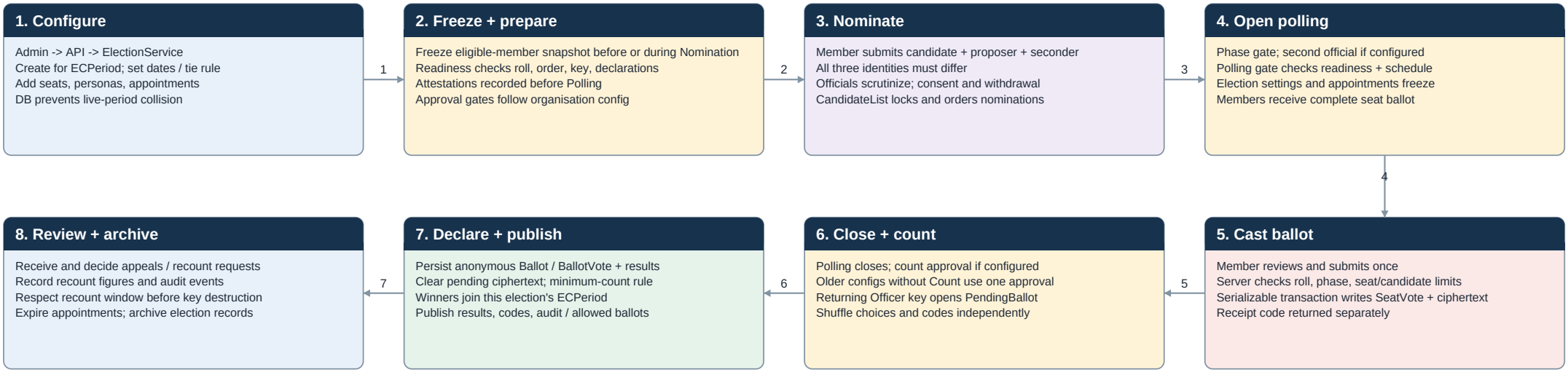
UML class diagram | Participation, ballots and outcomes

No identity-to-choice association. SeatVote stores a MemberId value and date only; the ERD shows the full electionId links.



Election workflow | Full lifecycle and control gates

End-to-end flow from election creation through appeals and archive. Approval is separate from execution.



Control gates and data transitions • Phase changes are validated by ElectionService. TwoPersonActions is organisation-configured; a distinct second official is required only for configured actions, subject to the SuperAdmin exception. Older configs without Count require one approval for counting. • VoterRoll may be frozen before or during Nomination. A vote is one full submission, while SeatVote proves participation separately and stores the vote date without time. • PendingBallot holds ciphertext only. At count, the service decrypts using the Returning Officer's supplied private key, shuffles ballots and receipt codes separately, writes anonymous counted records and removes pending rows. • Declaration writes winners into the EC period selected when the election was created. A failed readiness gate prevents Polling; a failed count leaves the election undeclared. Audit entries form a per-election hash chain.